We first note that $\bigvee_{j=1}^{n} p(i,j)$ asserts that row i contains atleast one queen,
and $\boxed{Q_1 = \bigwedge_{i=1}^{n} \bigvee_{j=1}^{n} p(i,j)}$ asserts that every row contains atleast one queen.

- For every row to include atmost one queen, it must be the case that $p(i,j)$ and $p(i,K)$ are not both true for integers j and K with $1 \leq j < K \leq n$. Observe that $\neg p(i,j) \vee \neg p(i,K)$ asserts that atleast one of $\neg p(i,j)$ and $\neg p(i,K)$ is true, which means that atleast one of $p(i,j)$ and $p(i,K)$ is false. So, to check that there is atmost one queen in each row, we assert

$$\boxed{Q_2 = \bigwedge_{i=1}^{n} \bigwedge_{j=1}^{n-1} \bigwedge_{K=j+1}^{n} (\neg p(i,j) \vee \neg p(i,K))}$$

To assert that no column contains more than one queen, we assert that

$$\boxed{Q_3 = \bigwedge_{j=1}^{n} \bigwedge_{i=1}^{n-1} \bigwedge_{K=i+1}^{n} (\neg p(i,j) \vee \neg p(K,j))}$$

(This assertion, together with the previous assertion that every row contains a queen, implies that every column contains a queen.)

To assert that no diagonal contains two queens, we assert

$$\boxed{Q_4 = \bigwedge_{i=2}^{n} \bigwedge_{j=1}^{n-1} \bigwedge_{K=1}^{\min(i+1,n-j)} (\neg p(i,j) \vee \neg p(i-K,K+j))}$$??

and

$$\boxed{Q_5 = \bigwedge_{i=1}^{n-1} \bigwedge_{j=1}^{n-1} \bigwedge_{K=1}^{\min(n-i,n-j)} (\neg p(i,j) \vee \neg p(i+K,j+K))}$$

The innermost conjunction in Q_4 and in Q_5 for a pair (i,j) runs through the positions on a diagonal that begin at (i,j) and runs rightward along this diagonal. The upper limits on these innermost conjunctions identify the last cell on the board on each diagonal.

- Putting all this together, we find that the solutions of the n-queens problem are given by the assignments of the truth values to the variables $p(i,j)$, $i = 1,2,\ldots,n$ and $j = 1,2,\ldots,n$ that make